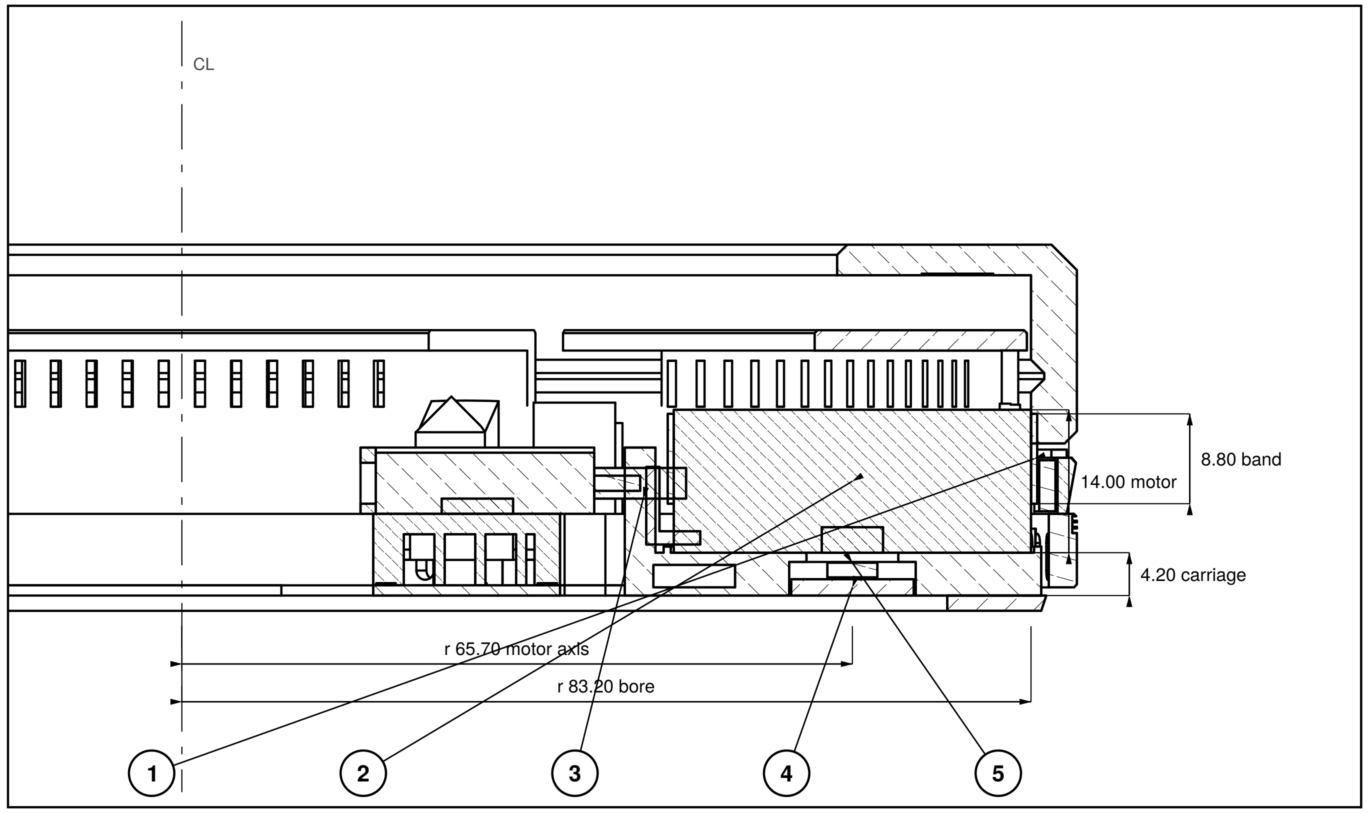
SECTION C-C scale 1.35 : 1



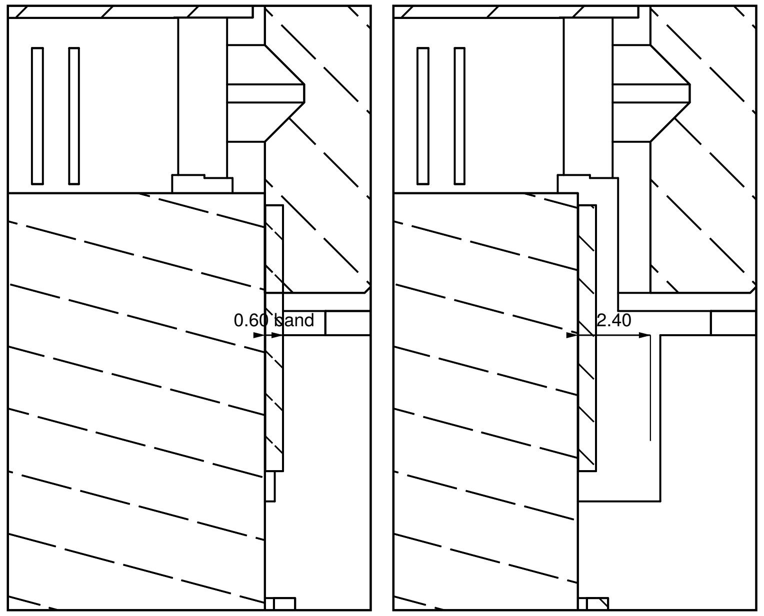
On the 90 deg - 270 deg axis, clutch ENGAGED. Right of the centreline is the motor at 90 deg; the servo's inboard end crosses the centreline.

NOTES

1. The drive is friction, not gears: a Ø35 motor bell running on the Ø166 bore through a 0.60 flat silicone band. Ratio 4.75 : 1. The gimbal drive and its over-centre clutch went at v9 and have not come back.
2. JD-Power MY-3514C (2804), dia 35 x 14, on a carriage that slides radially in a stadium hole through the aluminium core. The carriage shoe is dia 37 on the pad, so the motor's base face is at z 4.2 and its top at z 18.2.
3. The clutch IS the servo. An AGFRC C1.5CLS PRO linear servo (21.4 × 15.2 × 6 envelope) pushes the carriage tab directly; 2.40 of travel takes the bell from contact to 2.40 clear. Preload — 2.4 N nominal — is a servo position set in software, not a spring.
4. Commutation: an MT6701 on a 12 x 12 board inside the carriage, under the motor, reading a Ø6 × 2.5 diametric magnet in the shaft's lower end through a dia 9 sight hole. Which end of the shaft carries the magnet is ASSUMED. If it is the top, this board moves and the display rises about 3 mm, against 35.9 as modelled.
5. Not drawn, because nothing published says: the motor's bolt pattern (the carriage has a Ø35.6 locating rim only), the servo's mounting lugs, and the pushrod-to-tab joint. Measure all three on the real parts before printing the carriage or the servo mount.
6. Band grip at 2.4 N is untested. If it slips under a firm hand, the fallback is a 2 : 1 bell-crank between the servo and the tab — that doubles the force and halves the travel, so the stadium hole would grow.

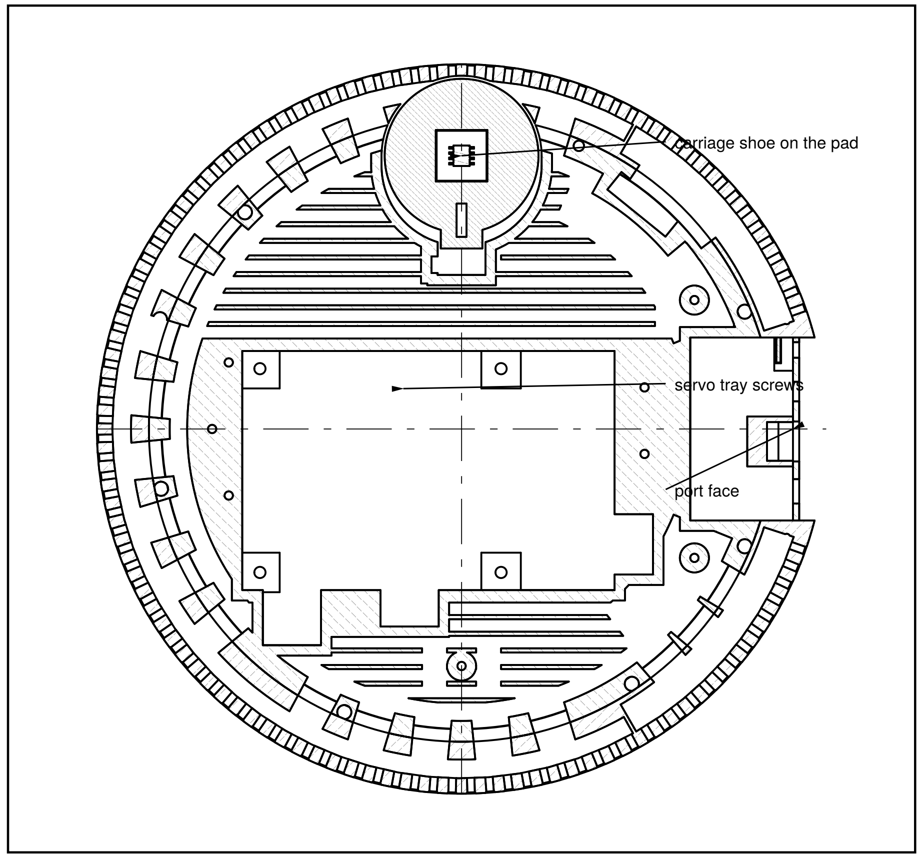
DETAIL R - THE CLUTCH 4 : 1

ENGAGED **RELEASED**



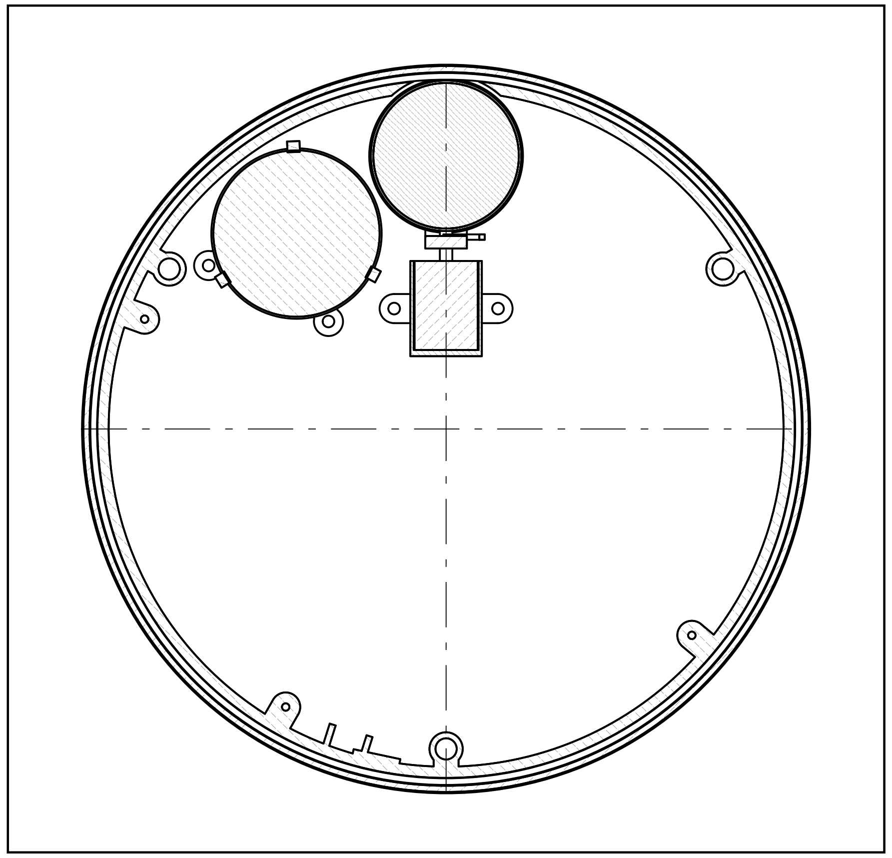
Same section, same scale, 2.40 apart. Engaged, the band is squeezed 0.60 between bell and bore. Released, the bell stands 2.40 clear and the knob free-spins. The knob is never released by the wheels — only by the drive.

PLAN AT z 2.5 - INSIDE THE CORE scale 0.55 : 1



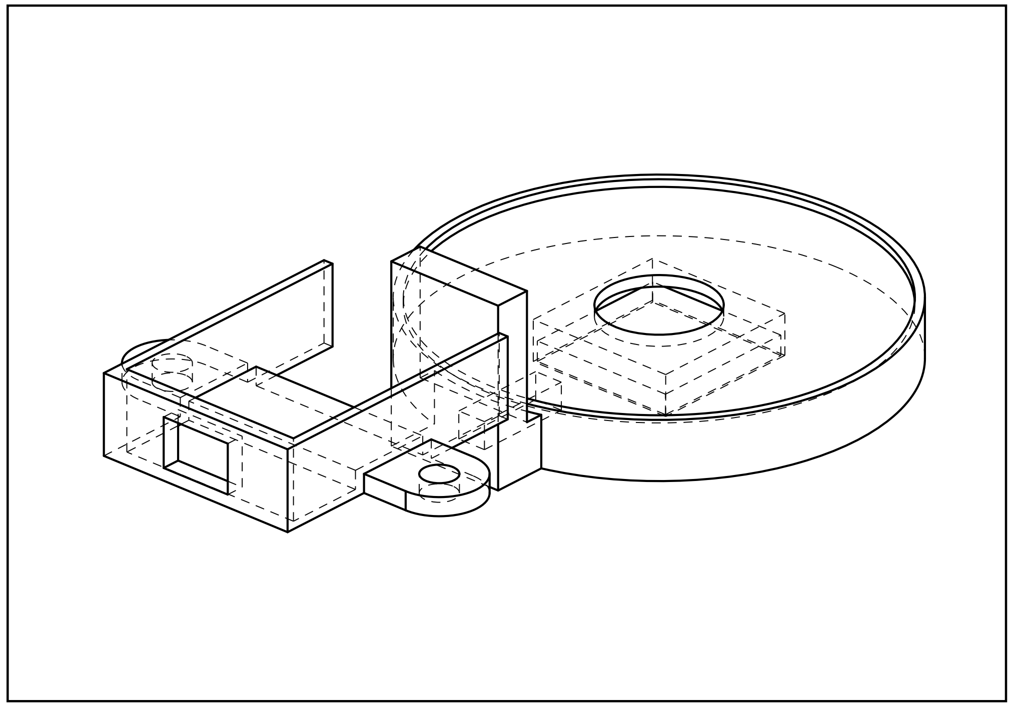
The stadium hole is the two carriage positions plus the slot between them: dia 38.5 wide, r 44.1 - 85.0 at 90 deg, with a 10 x 10 tab slot reaching in to r 39.7. The carriage rides on the pad, not on the core. This cut is inside the cooling duct - see D05.

PLAN AT z 13.4 - THE DRIVE BAND scale 0.55 : 1



Contact is a line at 90 deg, at r 83.20. Everything else on this level - motion board at 51 deg, speaker, servo tray - stays outside the bell. The wall relief is cut for the bell, 2.0 a side; the base below z 8.5 is narrower.

CARRIAGE AND SERVO TRAY - ISOMETRIC 1.9 : 1



The two printed parts of the drive, with the commutation board in its pocket. Motor and servo are left off: their envelopes hide everything that has to be made. Assembly: board into the carriage, motor on top, band over the bell, servo into the tray, tray on the plate, then set the preload in software.

DRIVE AND CLUTCH - CARRIAGE, MOTOR, BAND, SERVO

the 60 (Cadrane) — desk dial	DRAWING 60-15-D03	REV v15
SCALE AS SHOWN	DATE 2026-09-07	UNITS mm

MODELLED, NOT MADE — no v15 part has been produced. Dimensions are model values, not inspection results. Bodies tagged ENVELOPE or ASSUMED in the model are drawn as envelopes, not as the real part.

Projected from docs/v15/the60_v15_assembly.step by hidden-line projection, with dimension text written from docs/v15/the60_v15/src/params.py. No hand drafting. The knob is drawn on its turned profile: the knurl is on the part but is not drawn.